5 · Modelling

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5.1 Why one split is not a backtest

5.1.1 The textbook split

Definition 5.1 (Split)**.** A *split* cuts the sample by date into three contiguous blocks: **train**, on which coefficients are estimated; **validation**, on which choices between fitted models are made; and **test**, which is scored once and never chosen on.

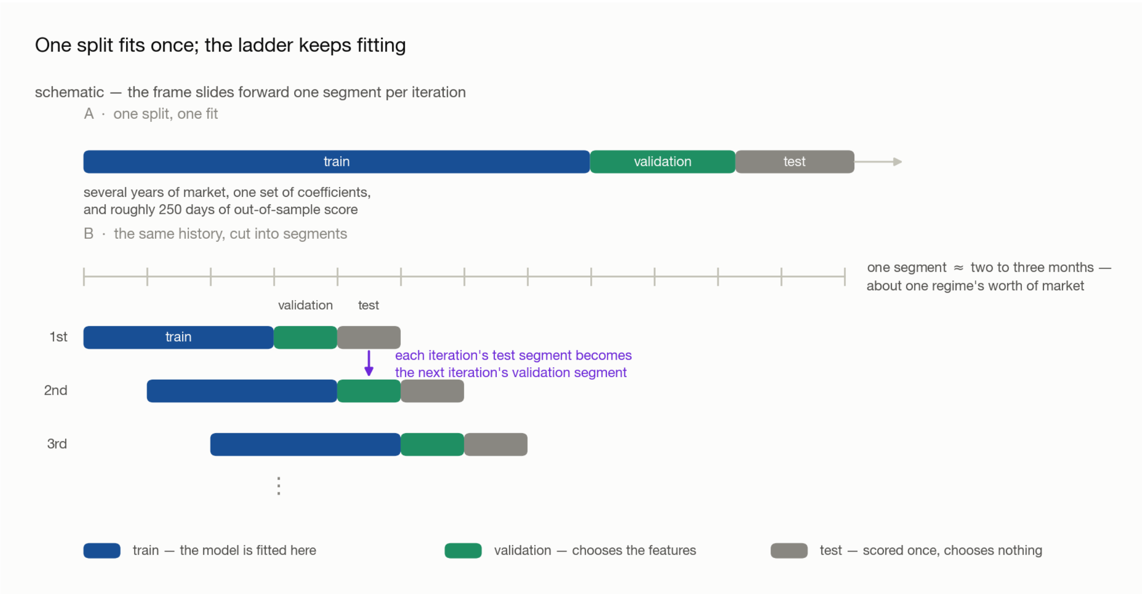

Figure 5.1: One split against the walk-forward ladder. Above, a single timeline cut once into train, validation and test. Below, the same history divided into twelve equal segments, with the train–validation–test frame starting one segment later on each row—so each row's test segment is the next row's validation segment.

The order matters and is not a convention: the blocks run forward in time, because a model that was fitted on next year and scored on last year has learned something no live book could have known.

5.1.2 What the single split costs

Three defects, and they compound.

Defect	What goes wrong
One fit discards time	A single set of coefficients over several years asserts that those years were one market. Early in the pandemic the world locked down at once and nobody wanted oil; the consensus was that demand had permanently moved online. It had not—the economy reopened, and energy then flew on war. A coefficient estimated across both states describes neither.
Almost nothing is left to score	Give validation a year and test a year and validation is roughly 250 dates. A year is also too little market: the regimes it contains are whichever ones happened to fall inside it, so the split never asks what the strategy does when the regime turns. A maximum drawdown measured over one calm year is not a risk estimate.
The sample is small and the signal is faint	A school dataset with a thousand points and an R^2 of 0.5 is ordinary, and in physics 0.9 is ordinary. In finance **0.1 to 0.2 is already a good** R^2—see 9. Weak true dependence plus few observations is precisely the regime in which a fit lands on noise: a line through ten points drawn from a cloud reaches R^2 0.8 easily.

5.1.3 The fix that is not one

Claim 5.2. Duplicating the sample changes no coefficient, and inflates every t-statistic by roughly $\sqrt{2}$.

Proof. Write the fit as $b = (X^T X)^{-1} X^T y$, with X the $n \times p$ matrix of signal values, y the forward returns, and b the estimated coefficients. Stacking the sample twice gives $X_2^T X_2 = 2\, X^T X$ and $X_2^T y_2 = 2\, X^T y$, so

$$b_2 = \left(2\, X^T X\right)^{-1} 2\, X^T y = b.$$

The residuals repeat as well, so the residual sum of squares doubles while the degrees of freedom go from $n - p$ to $2n - p$; for n large the variance estimate s^2 is therefore essentially unchanged. But the reported covariance is $s^2 (X_2^T X_2)^{-1} = s^2 (X^T X)^{-1}/2$ —halved. Standard errors fall by $\sqrt{2}$ and t-statistics rise by the same factor, on evidence that never arrived. □

Note. This is the standard interview question, and the reason it is asked is that people reach for it. Manufacturing rows is overfitting with extra steps; the answer to too few observations is to score more of the ones you have, which is §5.2.

5.2 The walk-forward ladder

5.2.1 How long a segment should be

Definition 5.3 (Segment)**.** A *segment* is a short contiguous block of dates—here **two to three months**—treated as one regime.

The length is set by how fast the market can change its mind, and the market is made of people. Within a day a regime shift is implausible: individuals change their view constantly, the crowd does not. Within a month, after a run of large news, one shift is plausible. Within a year or two there is room for several, because that is enough time for the same participants to change their minds repeatedly. So a segment must be short enough that one segment is approximately one regime, and long enough to hold a usable sample.

5.2.2 The slide

Cut the history into segments $1, 2, 3, \ldots$ and slide a frame across them:

Iteration	**Train**	**Validation**	**Test**
1	1–3	4	5
2	2–4	5	6
3	3–5	6	7

Claim 5.4. The ladder repairs all three defects of §5.1.2 at once.

Each fit now sees one regime's worth of data and is re-estimated as the market moves on, so time is no longer averaged away. Every segment past the first frame is eventually scored out of sample, so the out-of-sample record grows with the length of the history instead of being fixed at one year. And a regime shift that lands inside a validation segment is now a *test* the strategy has to pass, rather than an event the single split happened to miss.

Note. Read the table down its two right columns: **each iteration's test segment is the next iteration's validation segment.** Nothing is discarded between rungs, and no segment is ever scored twice under the same coefficients.

5.2.3 One rung, in full

A rung is not a single fit. Train ends, a validation segment intervenes, and only then does test begin—so a model fitted on train alone is asked to predict across a gap of market it has never seen, and pays for it.

The repair is the two-step shape that Lasso already imposes. Lasso adds an L_1 penalty to the least-squares objective,

$$\min_{\beta} \quad \sum_t \left(y_t - \sum_j \beta_j x_{j,t} \right)^2 + \lambda \sum_j |\beta_j|$$

where y_t is the forward return on date t, $x_{j,t}$ signal j's value on that date, β_j its coefficient, and $\lambda \geq 0$ the penalty strength. The penalty drives coefficients exactly to zero, which makes it a **feature selector**; and because the surviving coefficients are biased toward zero by that same penalty, the standard practice is to **refit** the survivors without the penalty before predicting.

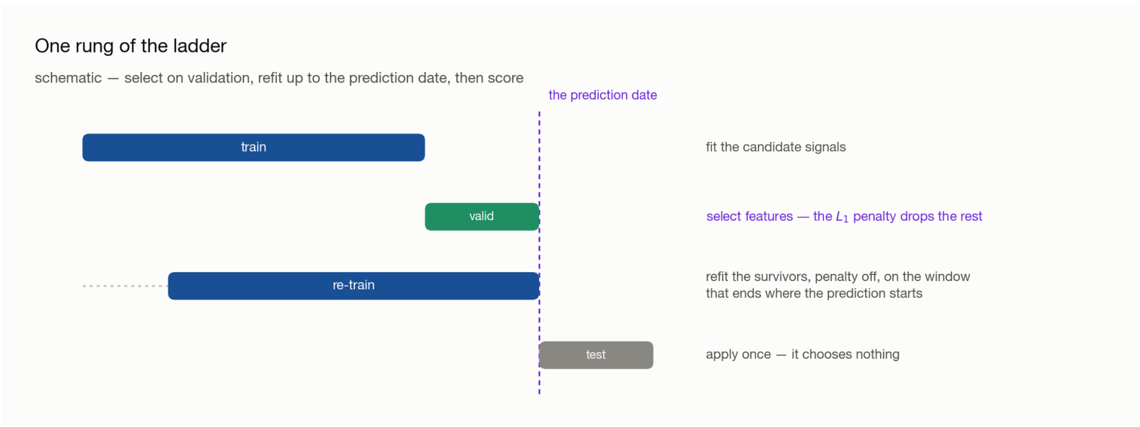

Figure 5.2: One iteration, in four stages: fit on train; select features on the validation segment; refit the survivors on a window shifted forward so that it ends where the prediction starts; apply once to test, beyond the dashed prediction date.

So each rung runs: **fit on train**, **select features on the validation segment**, **refit the survivors on a window ending where the prediction starts**, **apply once to test**.

Note (Two house styles)**.** The refit window is either the training window shifted forward, or the training window plus the validation segment. The sample sizes differ little and both are used in industry; it is a preference, not a correctness question. What is not optional is that the window ends at the prediction date—see §5.4.3.

5.3 Two validation layers

The validation *segment* inside each rung and the large held-out validation *block* at the end of the history are different objects with different powers.

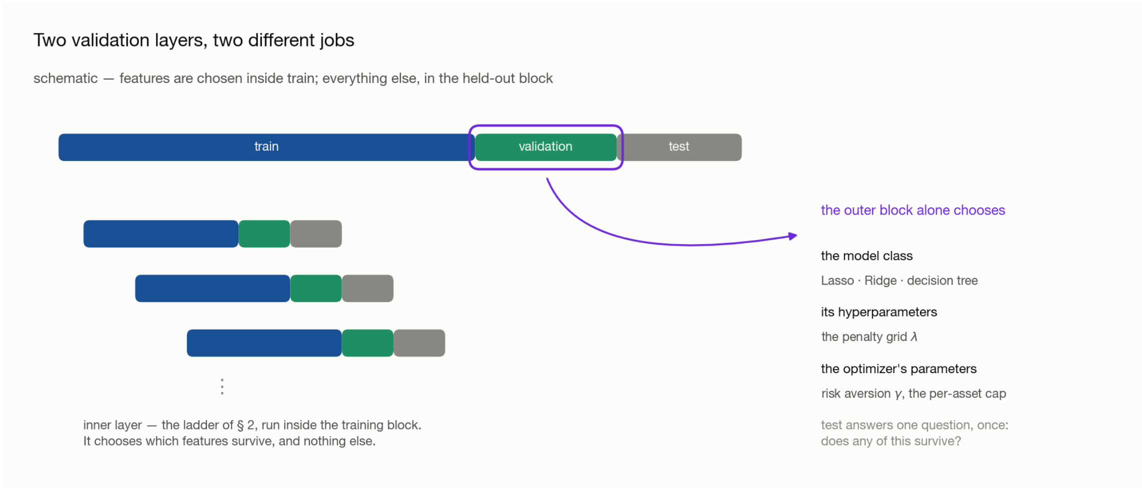

Figure 5.3: The two layers. The inner layer is the miniature ladder inside the training block, which chooses which features survive; the outer layer is the ringed held-out validation block, which chooses the model class, its hyperparameters, and the optimizer's parameters.

Layer	May choose	May not
Inner—the validation segment inside each rung	which **features** survive, and the refit that follows	anything held constant across rungs
Outer—the held-out validation block	the **model class**, its **hyperparameters** (λ), and the **optimizer's parameters**	coefficients—those are fitted on train
Test	nothing	everything

5.3.1 Why the inner layer cannot be skipped

Exploration already looked at which signals worked *in the training period*. Of course the fit then describes that period well: momentum was selected partly because it was seen to work there. That says nothing about whether it was useful going in, and the only way to find out is to select on dates the selection did not see—which is what the validation segment is for.

Note. A signal's usefulness is not a constant either. When more signals enter the book, or when one starts to fail, momentum should no longer carry the weight it carried alone. Because the ladder re-selects at every rung, that adjustment happens on its own rather than being asserted once.

5.3.2 What only the outer block may decide

λ is not fitted, it is searched: run the ladder once per value on a grid—0.1, 0.5, 0.7, 1—and compare. The same is true of the model class itself (Lasso, Ridge, a decision tree). Both comparisons need dates that no rung of the ladder consumed, which is exactly what the held-out block is.

5.3.3 The optimizer's parameters

The prediction is not yet a portfolio. Weights come from a mean–variance problem: choose w to

$$\max_{w} \quad E[r_p] - \gamma \cdot \mathrm{Var}[r_p]$$

subject to constraints such as

$$\sum_i w_i = 1.0 \qquad \sum_i |w_i| \leq 2.0 \qquad |w_i| \leq 0.20$$

—net exposure 100 percent, gross exposure at most 200 percent, and no single asset above 20 percent—where w_i is the weight on asset i, $E[r_p]$ and $\mathrm{Var}[r_p]$ the portfolio's expected return and variance, and $\gamma \geq 0$ the **risk-aversion coefficient**.

Two of these are free parameters, and neither can be reasoned to:

- **γ.** Nobody can say from first principles whether a risk appetite is 0.3, 0.5 or 1. The number has no interpretable units—it is a trade rate between two quantities measured in different things—so it is chosen by running the grid and reading the result.
- **The per-asset cap.** With 37 assets, a 20 percent cap may be too tight for the book to hold enough of whichever names are paying; 25 or 30 percent may be the right ceiling. Gross and net exposure, by contrast, are set by mandate and are not tuned.

Note. All of this tuning belongs in the **outer** block. Tuning γ inside the rungs would let a portfolio parameter be chosen on the same dates the features were chosen on, and the held-out block would then be measuring a strategy that had already been fitted to it.

5.4 Preparing a signal for a model

Two transformations stand between a raw signal and a model, and both exist to stop the fit from tracking things that are not the market.

5.4.1 Clipping: keep the observation, cap the value

Claim 5.5. A ranked signal does not fill its range evenly—the mass piles at both ends—and those ends move the fit far more than their count suggests.

The first half is empirical: momentum in particular pins at an extreme whenever a name runs, so the histogram is a barbell rather than a plateau. The second half is mechanical. In a least-squares fit each observation's pull on the slope grows with its distance from the mean of x, so a handful of far points can swing a coefficient that

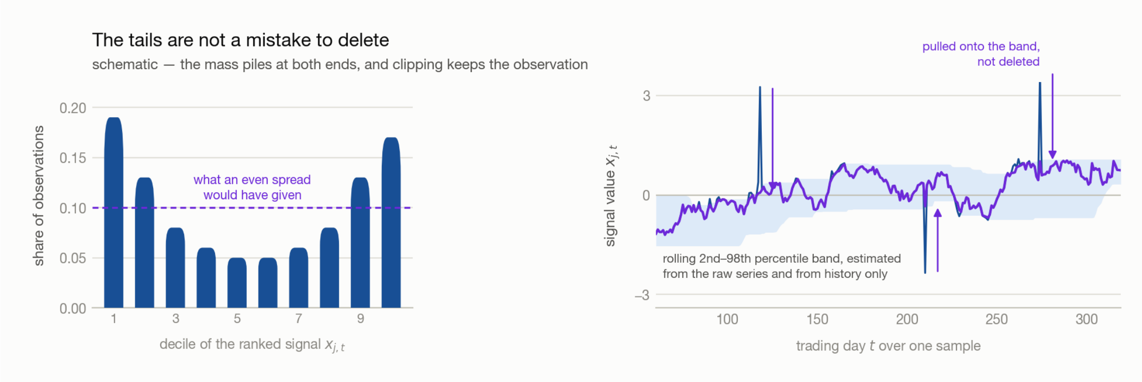

Figure 5.4: Left, the share of observations by decile of the ranked signal—high at the first and last deciles, low in the middle, against the dashed line at 0.1 that an even spread would give. Right, a signal series against a shaded rolling percentile band, with the spikes that reach outside it pulled back onto its edge.

hundreds of central points barely move. The coefficient then jumps for reasons that have nothing to do with the market.

Deleting the far points, which is the textbook move, fails twice here:

- **The sample is already small.** Every deletion sharpens §5.1.2's third defect, and in a messy sample it is not even clear which points are outliers.
- **The tail carries information.** What the book does when a signal goes extreme is exactly what the model should learn. Delete those dates and it never can.

Definition 5.6 (Winsorization)**.** Replacing any value outside a chosen quantile band by the nearer edge of the band:

$$x_{j,t}^{\mathrm{clip}} = \min\left(\max\left(x_{j,t},\ Q_{j,t}^{\mathrm{lo}}\right),\ Q_{j,t}^{\mathrm{hi}}\right)$$

where $Q_{j,t}^{\mathrm{lo}}$ and $Q_{j,t}^{\mathrm{hi}}$ are quantiles of signal j over the trailing L dates—1 and 99 percent, or 2.5 and 97.5 percent; there is no canonical pair.

Note (Compare raw to raw)**.** The band for date t must be estimated from the **unclipped** history. Feed yesterday's capped value into today's quantile and the band walks inward on itself, a little further every step. Keep the raw column and produce the clipped one beside it.

Note (Why the band rolls)**.** A value that is extreme against the last three months may be unremarkable six months later, once the market has moved to it. A fixed threshold pins such a signal at the 99th percentile forever; a rolling one releases it as soon as the surrounding distribution catches up.

→ `rolling`, `quantile` and `clip` as pandas calls: 8 · Toolbox.

5.4.2 Standardizing: put every signal on one scale

Momentum lives in roughly $[-0.05, 0.05]$. VIX lives in $[0, 100]$. Fit both and the coefficients absorb the mismatch.

Claim 5.7. Coefficient magnitudes carry no information about a signal's importance until the signals share a scale.

Proof. Replace x_j by $c\,x_j$ for some $c > 0$. The fitted values are unchanged if β_j is replaced by β_j/c, and least squares chooses fitted values—so the fit is identical and the coefficient is whatever the units make it. A comparison of β magnitudes across differently scaled signals is therefore a comparison of units. $\square$

The consequence is not merely aesthetic. Every input is an **estimate** and carries error: a momentum measured from returns is the residue of participants entering and leaving, and a jump in it is as likely to be noise as news. A large β multiplies that noise into the prediction, while the small β on the wide-ranging signal cannot respond when it genuinely moves.

Definition 5.8 (Standardized signal)**.**

$$z_{j,t} = \frac{x_{j,t} - \mu_j}{\sigma_j}$$

with μ_j and σ_j the mean and standard deviation of signal j over the estimation window. Dividing by a rolling standard deviation, or by a cross-sectional sum, are alternatives; what matters is that **every signal is treated the same way**.

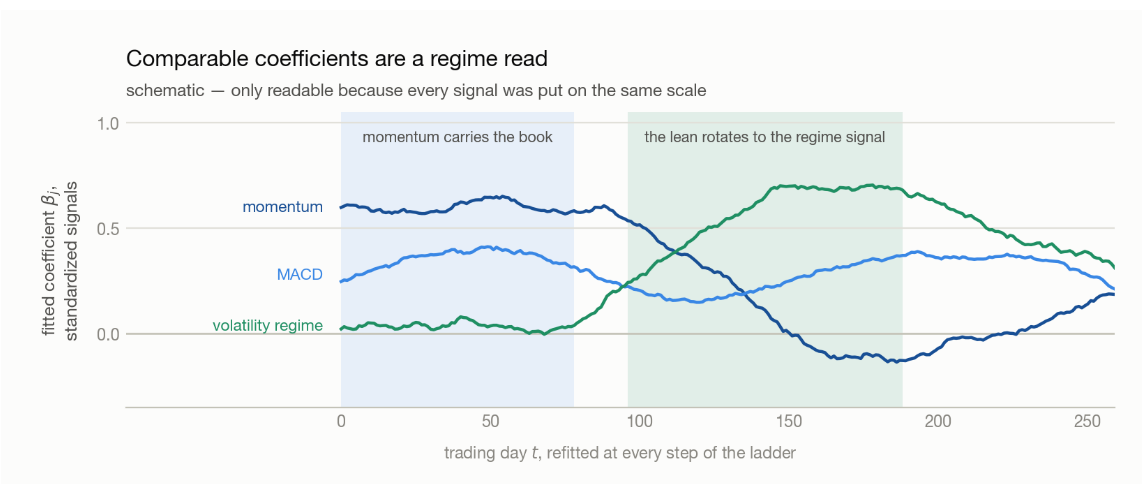

Figure 5.5: Three coefficient paths over a year of refits: momentum decays from high toward zero, a volatility-regime signal rises from zero to become the largest, and MACD stays between them. The shaded stretches mark where momentum carries the book and where the lean rotates to the regime signal.

Three things follow, in increasing order of usefulness:

Consequence	**Why**
Coefficients become comparable	With units removed, β_j measures response per standard deviation of signal—the same question for every signal.
The coefficient path becomes a regime read	Plot every β_j from every rung on one axis. Momentum dominating one stretch and volatility another **is** the change in market sentiment, in a form you can look at.
Errors partly cancel	On one scale, with comparable coefficients and errors that are close to independent across signals, the noise in the combined prediction averages down as signals are added, instead of being dominated by whichever signal has the largest units.

5.4.3 Which mean and standard deviation you may use

μ_j, σ_j and the quantile band all have to be estimated from somewhere, and that choice is where look-ahead enters.

The rule follows from **where the model is built**: at the *end* of the training window. Every date inside that window is history at that moment, so standardizing with the whole training window's mean and standard deviation is legitimate—even though, from the point of view of an individual date early in the window, the statistic used to scale it was computed partly from its own future.

At this point of construction	**These statistics are available**
End of the training window	the whole training window
End of the training window	**never** the validation segment, and never test
End of a refit window	the whole refit sample, validation segment included

Note. The distinction is between *a signal's* information set and *the model's*. A signal computed on date t may use only data up to t, because it has to be computable live. The model is not computed on date t; it is computed once, at the end of its window, and then applied forward. Confusing the two costs either a leak or an estimate worse than the one you were entitled to.

Note. This is the failure worth being paranoid about. In a time series the smallest leak makes the result beautiful, and the gap between a beautiful backtest and the live book is where the money goes.

Appendix · Notation

Throughout, t is the date, i indexes assets and j indexes signals.

Symbol	**Means**	**First used**
$x_{j,t}$, $x^{\text{clip}}_{j,t}$, $z_{j,t}$	signal j on date t: raw, after clipping, after standardizing	§5.2.3, §5.4.1, §5.4.2
y_t, β_j	the forward return being predicted, and the coefficient the fit puts on signal j	§5.2.3
X, y, b, n, p, s^2	the design matrix, the response, the estimated coefficients, the number of dates, the number of signals, the residual variance estimate	§5.1.3
λ	the L_1 penalty strength—searched on a grid, not fitted	§5.2.3
L	the trailing window, in dates, that the clip band is estimated over	§5.4.1
$Q^{\text{lo}}_{j,t}$, $Q^{\text{hi}}_{j,t}$	the low and high quantiles of that window	§5.4.1
μ_j, σ_j	the mean and standard deviation used to standardize signal j	§5.4.2
w_i, γ, $E[r_p]$	the portfolio weight on asset i, the risk-aversion coefficient, and the portfolio's expected return	§5.3.3

Note (Collisions to watch)**.** σ_j here is one **signal's** dispersion, used only to rescale it—not 2 · §4's $\sigma_{s,t}$, an asset's trailing volatility, and not 4 · §1.1's σ, the amplitude of the tape. β_j is a coefficient on a **signal**, as in 4 · §3.2, never a market beta. λ is the penalty and γ the risk aversion: the lecture wrote both as λ and then renamed the second, and they are unrelated. L is a trailing window as in 4 · §2.1, here over a signal rather than over returns.